# CSC Assignment 4 - Laptop Advisor Expert System

## Project Summary

This project is a rule-based laptop advisor that uses forward chaining to infer laptop recommendations and hardware specifications from user-provided facts. The system is written in Python and uses the 10 IF-THEN rules provided in Assignment 4_rules.docx.

## Program Structure

engine.py implements the inference engine. kb_loader.py reads rules from JSON. kb/laptop_rules.json contains the 10 laptop-advisor rules. main.py asks questions and prints results. tests/test_engine.py contains automated tests.

## Reasoning Process

1. The program collects budget, usage needs, portability, battery life, operating system preference, and other facts.
2. can_fire checks that every antecedent is known and that the conclusion is not already present.
3. run repeatedly selects an eligible rule (highest priority first), adds its consequent to working memory, and saves the rule in a trace.
4. When no more rules can fire, conclusions returns the recommendations and specifications. The trace explains each inferred result.

## Example

Input facts: budget_high, portable, long_battery.
Rule: IF budget_high AND portable AND long_battery THEN recommend:premium_ultrabook.
Output: premium_ultrabook. The explanation identifies the Premium Ultrabook rule.

## Testing and Results

The automated suite includes 13 tests covering all 10 provided rules, a no-match scenario, multiple antecedents, and a multi-step inference chain. All 13 tests passed. A sample interactive run also produced premium_ultrabook with the correct explanation.

## How to Run

Open the project folder in a terminal. Run `python main.py` to start the advisor. Run `python -m unittest discover -s tests -v` to execute the tests. Python 3.8 or later is required; no third-party

packages are needed.

## Limitations

The advisor only makes conclusions supported by the supplied rules. Some fact combinations produce specifications without a laptop recommendation, or no match at all. The rules are deterministic and do not assign confidence scores.